Example 10.33

In a Normal distribution 7% items are under 35 and 89% are under 63. Find its mean and standard deviation.

Solution:

$$Z = \frac{x - \mu}{\sigma}$$

We have $P(x < 35) = 7\% = 0.07$

If $z_1 =$ then $P(z_1 < Z < 0) = 0.50 - 0.07 = 0.43$

$\therefore z_1 = -1.48$ (from Normal Probability table)

$$\frac{35 - \mu}{\sigma} = -1.48$$

$$35 - \mu = -1.48\sigma \quad \ldots (1)$$

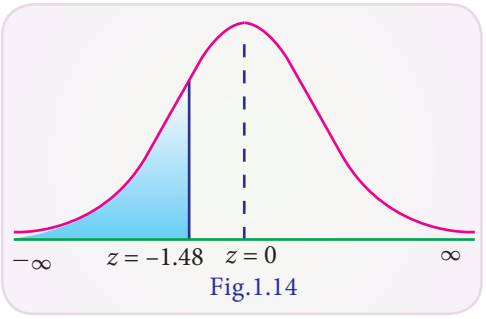


Fig.1.14

Also $P(X < 63) = 89\% = 0.89$

If $Z_2 =$ then $P(0 < Z < Z_2) = 0.89 - 0.50 = 0.39$

$Z_2 = 1.23$ (from Normal Probability table)

$$\frac{63 - \mu}{\sigma} = 1.23 \text{ (or) } 63 - \mu = 1.23\sigma \quad \ldots (2)$$

$$(2) - (1) \Rightarrow 28 = 2.71\sigma \text{ or } \sigma = \frac{28}{2.71} = 10.33$$

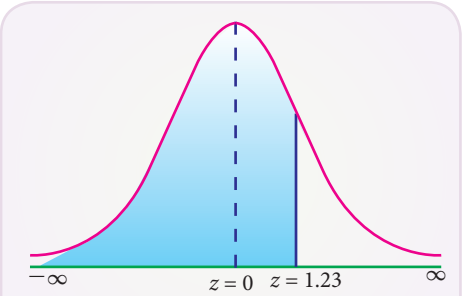


Fig.10.15

From (1) wehave

$$\mu = 35 + 1.48\sigma = 35 + 1.48\ (10.33)$$

$$\mu = 50.27$$

10.3 Fitting of Binomial, Poisson and Normal distributions

Introduction

Fitting of probability distribution to a series of observed data helps to predict the probability or to forecast the frequency of occurrence of the required variable in a certain desired interval.

There are many probability distributions of which some can be fitted more closely to the observed frequency of the data than others, depending on the characteristics of the variables. Therefore one needs to select a distribution that suits the data well.